
From PDE Residuals to Error Rankings in Elliptic Neural Operators

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Abstract

1 A PDE residual checks whether a neural operator’s prediction satisfies the equation.
2 But does it also show where the prediction is wrong? On elliptic problems across
3 two grids and three binary coefficient contrasts, and in a separately trained smooth-
4 coefficient control, raw residual maps locate pointwise error poorly, below simple
5 prediction and geometry scores. Partial solves recover much better error maps. In
6 one challenging setting, estimated and true top-10% error sets overlap by 99.15%,
7 yet a classical bound certifies only 16.3% of the true set as members. Most of the
8 prediction correction is already computed by this stage. Checking the equation,
9 ranking spatial errors and certifying which cells have the largest errors are distinct
10 tasks; moving between them has a solve cost.

11 1 Introduction

12 Neural operators learn to predict PDE solutions from data [Li et al., 2021]. How can we assess a new
13 prediction without a reference solution? Substituting it into the equation gives a residual: a check of
14 equation consistency, which can itself be a target for uncertainty quantification [Gopakumar et al.,
15 2025]. Locating large prediction errors asks more. A cell where the equation is poorly satisfied need
16 not be a cell where the solution error is large.

17 To see the distinction, consider the discrete problem $Au = b$. For a prediction $\hat{u}$, write $e = \hat{u} - u$ for
18 its error and $r = A\hat{u} - b$ for its residual. Then

$$r = Ae, \quad e = A^{-1}r. \quad (1)$$

19 For diffusion, the residual measures the divergence of an error flux. Nearby face contributions can
20 cancel, and their weights depend on the material coefficient. Its pattern therefore differs from the
21 error itself. Applying A^{-1} recovers the error but requires another solve. A useful diagnostic must
22 balance the information it provides against this work.

23 The identity $e = A^{-1}r$ is classical. Our contribution is to measure what information about a
24 learned predictor’s spatial error becomes available at intermediate solve budgets, before the inverse is
25 fully applied. We compare equation consistency, empirical ranking, certified top-error membership
26 and actual prediction correction. Our tools build on preconditioning and error estimation [Li and
27 Zikatanov, 2021], neural-operator residual correction [Cao et al., 2022, Jha, 2024], and pointwise and
28 equilibrated-flux bounds [Anwer et al., 2026, Ern and Vohralík, 2011]. With simple baselines, fresh
29 fields and measured solve costs, we show that accurate rankings can support few certified decisions
30 while already computing most of the prediction correction.

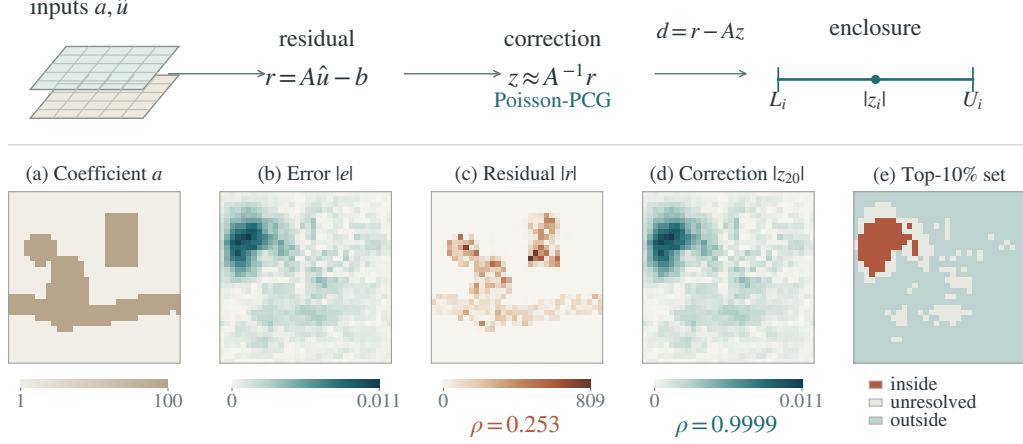


Figure 1: Error maps and conservative decisions for one fresh 32^2 , $\kappa = 100$ field, selected by proximity to median raw-residual Spearman. Reference error $|e|$ is used only for evaluation and shares a color scale with $|z_{20}|$. The flux bound identifies members and nonmembers of the top-10% error set, leaving others unresolved. Aggregate results use all 200 fields.

2 Experimental setup

Problem and predictions. We solve $-\nabla \cdot (a \nabla u) = f$ on $[0, 1]^2$, with zero Dirichlet boundary conditions and $f(x, y) = 1 + 0.25 \sin(2\pi x) \sin(\pi y)$. The coefficient takes values $a \in \{1, \kappa\}$ in random inclusions and channels; $\kappa \in \{10, 100, 500\}$ controls the contrast. We use 32^2 and 64^2 cell grids. A cell-centered two-point-flux discretization, with harmonic face coefficients and half-cell boundary distances, gives a symmetric positive definite matrix A .

Each grid and contrast has a separate ensemble of three Fourier neural operators (FNOs): 700 training fields, 80 epochs, three layers, 12 Fourier modes and 24 channels per member. Normalization uses training data only; $\hat{u}$ is the ensemble mean. After exploration on 200 original test fields, we fixed methods and budgets, then drew 200 fresh geometries matched across contrasts and grids. Models remain frozen; resolutions are trained separately. Predictions use float32; solves and diagnostics use float64. Eq. (1) holds with the same A to relative tolerance 10^{-8} on every field. Bounds and training controls are subsequent exploratory checks; optimization settings are in the artifact.

A matched smooth-family FNO check uses 32^2 , $\kappa = 100$, the same forcing and training budget, and independent 700/200 train/test fields. We Gaussian-filter white noise (standard deviation 0.12 domain lengths), scale each field to $t \in [0, 1]$, and set $a = 100^t$. Comparisons with binary media are descriptive: the coefficient distribution and trained predictor both change.

Comparisons and measurements. Cheap scores are ensemble standard deviation, prediction magnitude $|\hat{u}|$, distance to the nearest boundary, and residual magnitude $|r|$. Partial solves of $Az = r$ use $|z|$ to estimate spatial error. We compare undamped Jacobi, conjugate gradients (CG), diagonal-PCG, and one full-hierarchy smoothed-aggregation algebraic multigrid (AMG) V-cycle [Bell et al., 2023]. The Poisson matrix $B = A(1)$ is inverted by an orthonormal discrete sine transform matched to the cell boundaries. We use this inverse directly or as a CG preconditioner (Poisson-PCG). Iterations start from zero, with budgets of 5, 10, 20, 40, 80 steps.

We average spatial Spearman correlations with $|e|$ over fields. Percentile 95% intervals use 5,000 field-bootstrap resamples, paired across methods and conditional on the frozen networks. The top 10% is a representative hotspot budget; ties are treated uniformly. We also compare cell error energies: nonnegative face contributions, including boundaries, that sum to $e^T A e$.

Turning an estimate into a decision. An error map alone does not justify each cell's ranking. We bound the error left after a correction z , writing $d = r - Az$ and $a_{\min} = \min_i a_i$. Since $A \succeq a_{\min} B$

in quadratic-form order, a classical pointwise bound [Anwer et al., 2026] gives

$$|e_i - z_i| \leq \beta_i := \frac{\sqrt{(B^{-1})_{ii} d^T B^{-1} d}}{a_{\min}}. \quad (2)$$

Energy Cauchy–Schwarz and inverse ordering give this bound. It adds one Poisson solve, with the inverse diagonal precomputed per grid; the artifact details the derivation and setup costs.

A tighter bound uses the actual face coefficients, following equilibrated-flux estimation [Ern and Vohralík, 2011]. Write $A = DW_A D^T$ and $B = DW_B D^T$, with cell-face incidence D and diagonal face weights, including boundaries. The Poisson flow $q = W_B D^T B^{-1} d$ satisfies $Dq = d$ and bounds the remaining error energy by $E_{\text{eq}} = q^T W_A^{-1} q$. This gives

$$|e_i - z_i| \leq \beta_i^{\text{eq}} := \sqrt{(B^{-1})_{ii} E_{\text{eq}} / a_{\min}} \leq \beta_i. \quad (3)$$

This version reuses the Poisson solve and adds coefficient-weighted face sums.

Either bound, denoted by $\bar{\beta}_i$, gives an interval for the error magnitude: $L_i = \max(|z_i| - \bar{\beta}_i, 0)$ and $U_i = |z_i| + \bar{\beta}_i$. We certify a cell as belonging to the largest-error set when its lower bound exceeds enough of the other upper bounds. Specifically, for $m = \lceil 0.1N^2 \rceil$, it is certainly inside if $L_i > U_{(m+1)}$, and certainly outside if $U_i < L_{(m)}$, with parentheses denoting descending order. Other cells remain unresolved; strict inequalities handle ties. These are exact-arithmetic implications for the discrete problem, not interval-arithmetic guarantees for our float64 checks. Ground truth only evaluates the decisions. An exploratory rule checks every five iterations, stopping at $\lceil 0.9m \rceil$ identified members or 80 iterations. With valid bounds, reaching this target guarantees 90% top-set recall without a reference solution.

3 Results

Table 1: Mean spatial Spearman with $|e|$ on fresh 32^2 binary fields, with 95% field-bootstrap intervals. Equal iterations are not equal work; 1.000 denotes rounding, not exact recovery.

Score / correction	$\kappa = 10$	$\kappa = 100$	$\kappa = 500$
raw $ r $	0.276 [0.264, 0.289]	0.244 [0.227, 0.261]	0.214 [0.194, 0.233]
ensemble σ	0.238 [0.223, 0.253]	0.318 [0.302, 0.333]	0.327 [0.311, 0.343]
prediction $ \hat{u} $	0.411 [0.385, 0.438]	0.448 [0.425, 0.471]	0.441 [0.417, 0.464]
boundary distance	0.412 [0.387, 0.437]	0.410 [0.388, 0.432]	0.368 [0.344, 0.393]
Jacobi-20	0.489 [0.469, 0.508]	0.508 [0.491, 0.524]	0.531 [0.513, 0.550]
CG-20	0.551 [0.532, 0.569]	0.355 [0.344, 0.368]	0.278 [0.263, 0.293]
diagonal PCG-20	0.786 [0.768, 0.803]	0.681 [0.658, 0.703]	0.670 [0.646, 0.695]
$ B^{-1}r $	0.613 [0.594, 0.632]	0.408 [0.384, 0.431]	0.355 [0.330, 0.381]
one AMG V-cycle	0.986 [0.984, 0.987]	0.920 [0.907, 0.932]	0.867 [0.847, 0.885]
Poisson-PCG-20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.999 [0.999, 1.000]

Residual maps miss much of the pointwise error pattern. As Table 1 shows, raw residuals give weak rankings, below even prediction magnitude and distance to the boundary. They are more closely related to error energy. At 64^2 and $\kappa = 500$, for example, the mean residual correlation is 0.118 with pointwise error and 0.731 with energy. This is an empirical finding, not a consequence of Eq. (1) alone.

The mismatch persists with more accurate predictors (Table 2). Longer FNO training and a U-Net ensemble [Ronneberger et al., 2015] reduce prediction error without improving raw-residual ranking. The U-Net uses the same fields, three seeds, 200 epochs and two levels with 16/32/64 channels.

The mismatch also occurs without binary interfaces (Table 2). For smooth fields, twenty Poisson-PCG steps give top-set overlap of 99.5%, yet the flux bound certifies only 36.5% of the true top set as members (CI [33.9, 39.1]%). Only 0.59% of the original prediction error remains in L^2 .

Partial solves recover useful error maps. Preconditioning improves rankings at fixed iteration budgets (Table 1), with further gains from more iterations (Fig. 2a). Both AMG and Poisson-PCG are effective: strong localization is possible with or without a variable-coefficient hierarchy.

Table 2: Controls at 32^2 , $\kappa = 100$: means and 95% field-bootstrap intervals on 200 fields per row. The smooth FNO is separately trained; PCG-20 uses the Poisson preconditioner.

Predictor (epochs)	Rel. L^2 error (%)	Residual ρ	PCG-20 ρ
Binary FNO (80)	10.54 [9.44, 11.75]	0.244 [0.227, 0.261]	0.99998 [0.99998, 0.99999]
Binary FNO (240)	9.03 [8.00, 10.16]	0.223 [0.204, 0.240]	0.99999 [0.99998, 0.99999]
Binary U-Net (200)	4.41 [3.75, 5.15]	0.231 [0.211, 0.250]	0.99997 [0.99996, 0.99998]
Smooth FNO (80)	4.13 [3.92, 4.36]	-0.041 [-0.062, -0.020]	0.99994 [0.99994, 0.99995]

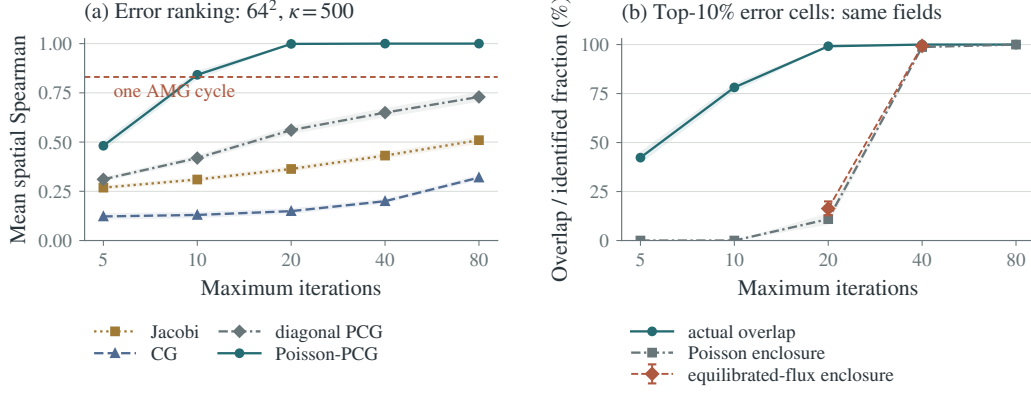


Figure 2: Ranking and certification at 64^2 , $\kappa = 500$ on fresh binary fields, with 95% field-bootstrap intervals. (a) Rankings improve with budget; AMG uses one cycle. (b) High top-10% overlap can coexist with few certified members. Flux bounds use 20/40 iterations. Each bound evaluation adds one Poisson solve; equal iteration counts are not equal costs.

93 **Accurate rankings can still leave most decisions unresolved.** At 64^2 , $\kappa = 500$, twenty Poisson-
94 PCG steps give a correlation of 0.9985. The estimated and true top-10% sets overlap by 99.15%
95 (CI [98.91, 99.34]%). Yet the Poisson bound certifies only 10.9% of the true set as members (CI
96 [8.0, 13.9]%). The flux bound raises this fraction to 16.3% (CI [12.9, 20.0]%).

97 With forty iterations, the Poisson bound identifies 98.7% of the true set (CI [98.4, 99.0]%). The
98 adaptive rule reaches its 90% target on all 1,200 fresh binary fields. We observe no false inside/outside
99 flags across the tested fields and budgets. Numerical enclosure checks allow $10^{-10}\|e\|_\infty + 10^{-13}$
100 floating-point slack; this slack is not added to the bounds used for decisions.

101 **Excellent localization already computes much of the correction.** Twenty Poisson-PCG steps
102 at 64^2 , $\kappa = 500$ leave only 1.29% of the initial error (mean $\|z - e\|_2 / \|e\|_2$) and also supply an
103 improved prediction $\hat{u} - z$. Assessing the diagnostic must include this correction.

104 For binary media, we time diagnostics on 30 fixed fresh fields per setting, using three repetitions,
105 randomized method order and one Apple M4 CPU thread. At $\kappa = 500$, the median paired Poisson-
106 PCG-20/direct-solve time ratio is 0.325 at 64^2 (CI [0.319, 0.328]) and 0.770 at 32^2 . Given A and
107 $\hat{u}$, timings include residual formation and amortize reusable Poisson setup; they do not establish
108 end-to-end speedup. The flux bound reduces adaptive iterations yet costs 1.116 times the original
109 rule at 64^2 : extra face checks offset the saving.

110 4 Discussion and reproducibility

111 These discrete results cover one forcing, two modest grids and compact predictors. Continuum
112 bounds, broader forcing families and universal contrast trends are not established.

113 Per-field records generate the tables and figures. Checks cover operator identities, independent inverse
114 calculations, energies, interval separation and frozen-model replay. Codex-assisted code was subject
115 to these checks; scientific design, analysis, and interpretation were author-directed. The anonymous
116 artifact contains code, seeds, checkpoints and records.

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